VECTOR

vPblCalib

Technical Reference

Version 1.1.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
Yves Grau, Steffen Köhler	2018-01-12	1.00.00	Initial Version
Yves Grau	2018-03-26	1.01.00	Added configuration section

Reference Documents

No.	Source	Title	Version
[1]	ASAM e.V.	ASAM_AE_MCD-2_MC_BS_V1-7-0.pdf	1.7.0
[2]	ASAM e.V	ASAM_AE_CDF_BS_V2-1-0_Reference_Guide.pdf	2.1.0
[3]	ASAM e.V.	ASAM_AE_CDF_BS_V2-1-0_Users_Guide.pdf	2.1.0
[4]	Vector	DVCfg_AutomationInterfaceDocumentation.pdf	See delivery
[5]	Vector	TechnicalReference_PostBuildXmlGenerator.pdf	See delivery
[6]	Vector	ReferenceManual_HexView.pdf	See delivery



Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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1 Component History

The component history gives an overview over the important milestones that are supported in the different versions of the component.

Component Version	New Features
1.00.00	Initial Version
1.01.00	Added support for configurable virtual memory segment start address

Table 1-1 Component history

2 Introduction

This document describes the functionality, API and configuration of the component vPblCalib. vPblCalib allows Post-Build modification of MICROSAR BSW parameters via a standard calibration tool which supports the transformer concept of ASAM MCD-2 MC version 1.7.0 [1].

During calibration, the configuration of basic software modules such as DEM and FiM may need to be changed. As these configurations are represented in memory in a compressed format, any direct changes via standard calibration tools are not possible. Prior to the introduction of vPblCalib, using DaVinci Configurator 5 was the only way to change parameters of software modules.

vPblCalib enables the offline calibration of parameters in MICROSAR modules via standard calibration tools such as CANape or vCDMStudio. Thereto, it performs the transformation between the compressed data representation of the binary in an ECU and the uncompressed physical representation required by the calibration tool. The calibration tool can then be used to change the value of a parameter which triggers vPblCalib that interacts with DaVinci Configurator 5 to execute the Post-Build Loadable process in background with no user interaction needed. The result is in an updated Post-Build Loadable section within the calibration tool.

2.1 Architecture Overview

The vPblCalib relies on the transformer concept introduced with ASAM MCD-2 MC version 1.7.0 [1]. Transformers as specified in [1] offer a standardized interface. This interface offers an additional way of converting parameter values from their ECU-internal format to a physical representation shown in the calibration tool. The exchange format is the ASAM Calibration Data Format (CDF) version 2.1.0 specified in [2] and [3].

Figure 2-1 shows the interfaces to adjacent tools of the vPblCalib. These interfaces are described in more detail in chapter 5.

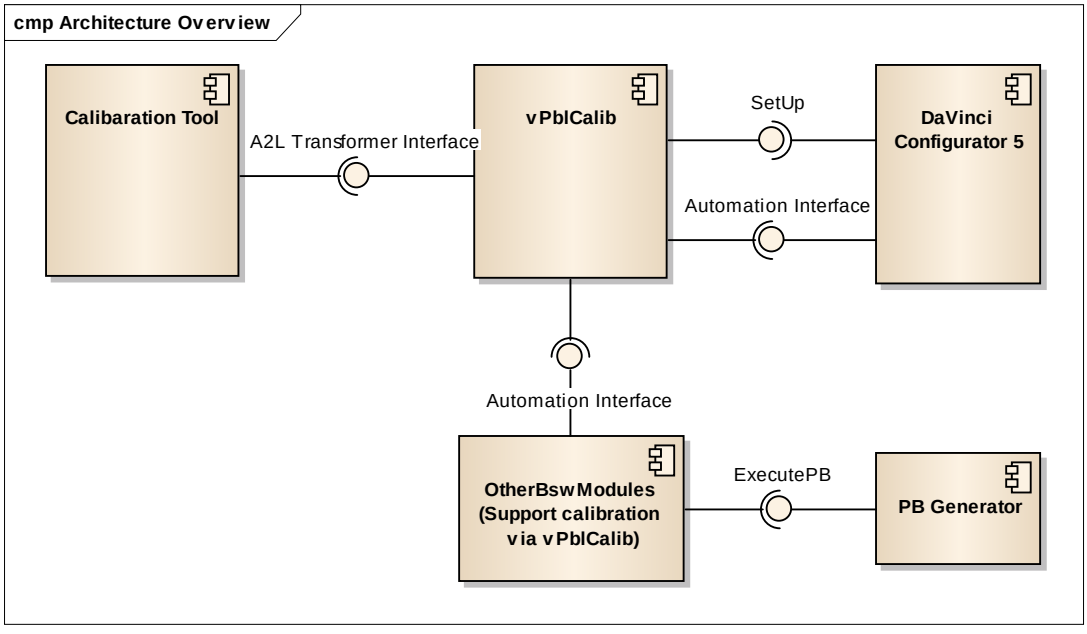


Figure 2-1 Interfaces to adjacent tools of the vPblCalib

3 Functional Description

3.1 Features

The features listed in Table 3-1 cover the complete functionality specified for the vPblCalib.

Supported Features
Calibration of MICROSAR modules that support this process
Calibration tool independency: Use standardized A2L transformer interface specified in [1]
Integration of customized Post-Build Loadable tool chain

Table 3-1 Supported features

3.2 Workflow

The workflow of vPblCalib can be divided into two phases. The Setup phase where the initialization for later calibration takes place and the actual Calibration phase where standard calibration tools trigger changes of BSW parameters.

3.2.1 Setup Phase

Figure 3-1 illustrates the vPblCalib workflow in Setup phase. For initialization, the vPblCalib offers an additional generation step in the generation view of Cfg5. Result is an A2L- and CDF-file. The A2L-file defines all BSW parameters enabled for calibration depending on the actual configuration. In addition, the CDF-file represents the concrete parameter data set of the calibratable parameters. The created CDF-file is stored in a collection of parameter data sets which is needed in *Calibration Phase*. The generated A2L-file needs to be merged into the A2L-file of the calibration process. Input for the standard calibration tool are the master A2L-file in combination with the HEX-file created by the build process.

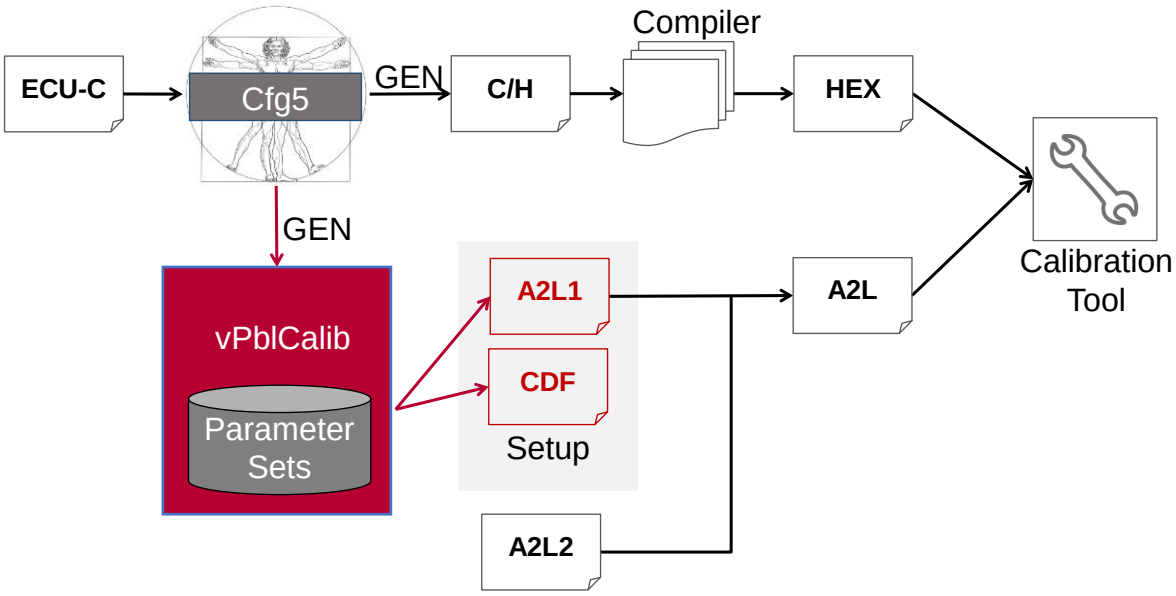


Figure 3-1 Workflow overview of the Setup phase

3.2.2 Calibration Phase

vPblCalib offers two A2L transformers. One transformer converts the compressed ECU representation available in the HEX-file to the uncompressed physical representation shown in the calibration tool. The second transformer is doing the conversion vice versa.

Figure 3-2 shows the workflow of BSW parameter decompression during calibration phase. The interaction between vPblCalib and the standard calibration tool is realized by the standardized A2L Transformer Interface. For the decompression of the BSW parameters the vPblCalib is not directly interpreting the compressed data. It is rather resolving a mapping between the compressed and uncompressed data via a collection of parameter sets created in Setup and Calibration phase. The uncompressed data is written to a virtual memory section within the HEX file which is then read and evaluated by the calibration tool.

Caution
The calibration or flash tool is responsible to only flash the memory segments that contain the compressed data and especially to not flash any part of the virtual memory section.

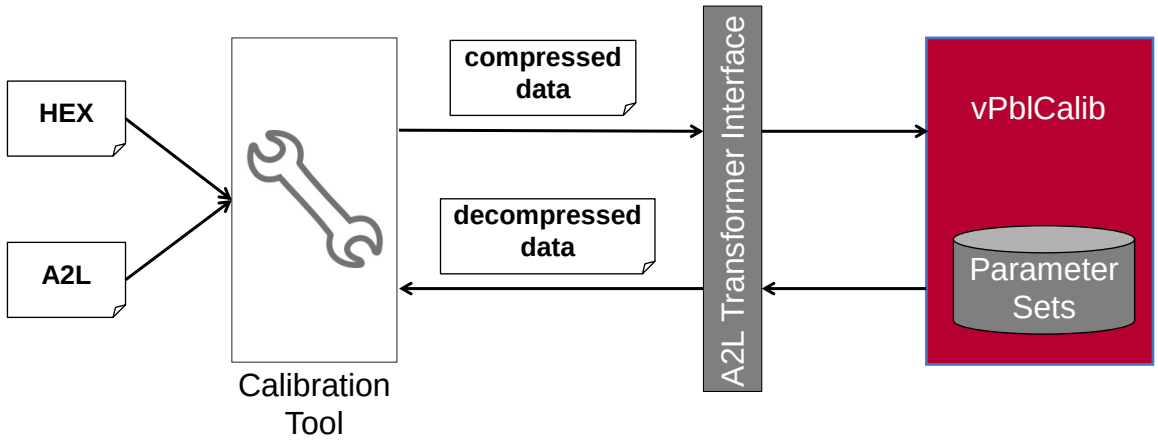


Figure 3-2 Decompression of BSW parameter values in Calibration phase

The workflow during Calibration phase of BSW parameter compression is shown in Figure 3-3. Same as for the decompression is the interaction between vPblCalib and the calibration tool done with the standardized A2L Transformer Interface. vPblCalib compresses the data with the conventional Post-Build Loadable feature of the Cfg5. First, the data model of Cfg5 is updated according to the parameter changes done in the calibration tool. Second, the Cfg5 generates the new C/H files. Third, the Post-Build generation is triggered via the PB Generation Interface which allows triggering PB tool chain of Vector or any user defined tool chain as well as a concrete compiler.

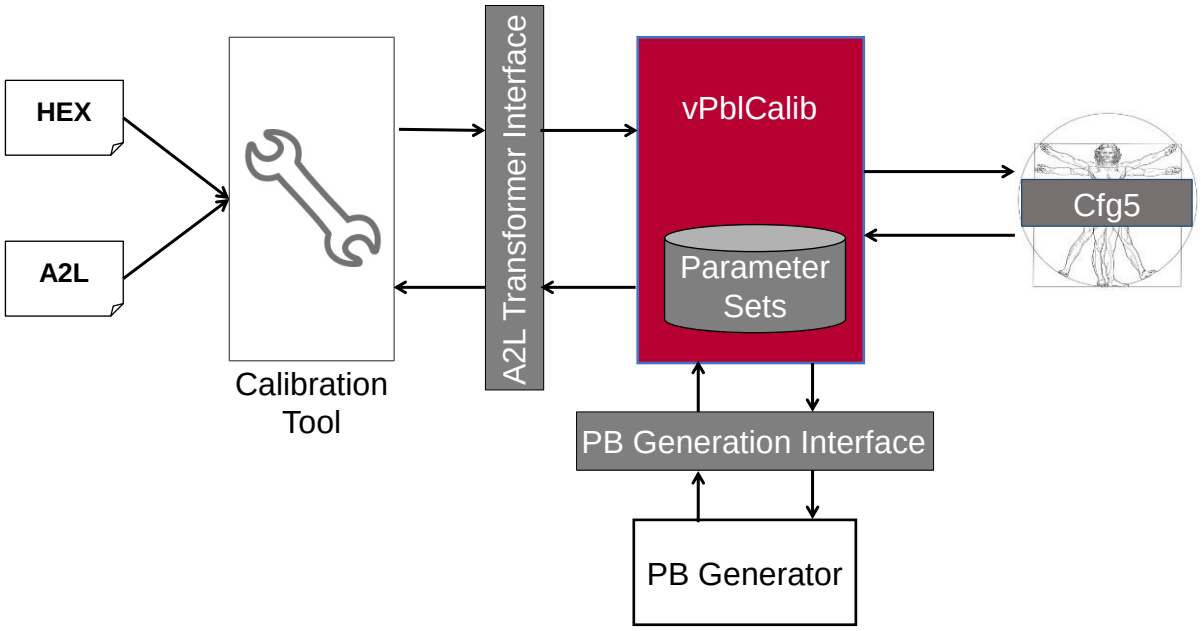


Figure 3-3 Writing of BSW parameter values in Calibration phase

3.3 Limitations

vPblCalib only supports offline calibration. It is not possible to change parameters in the running ECU (online calibration) via vPblCalib. Calibration parameters are represented in the ECU using compressed and optimized data structures that do not allow to make specific changes to single parameter values. Therefore, changing a parameter value requires to execute the whole Post-Build Loadable tool chain and to re-flash the resulting PB Loadable section of the ECU.

vPblCalib can be used with all calibration tools that support the A2L transformer concept specified in ASAM MCD-2 MC version 1.7.0 [1]. Current versions of the Vector tools CANape and vCDMStudio 16.0 support this version of the ASAM standard and can be used to work with vPblCalib.

vPblCalib supports calibration of most PB Loadable parameters of calibratable MICROSAR modules. Calibration of PBS parameters and creation of optional parameters during the calibration phase is not possible.

During Calibration Phase, vPblCalib supports only SIP paths which are free of spaces. Essential is the CommonAsr_vPblCalib.config file, for more details see 4.1.3.

3.4 Logging

vPblCalib offers a logging mechanism to support the integration process. To enable the logging mechanism create a file “logging.properties” in the SIP directory of the vPblCalib (“<SIP_Path>/Misc/vPblCalib/”). Figure 3-4 shows the relevant content of the file and the available properties of the logging mechanism.

```
# Here the location of the logfile is specified
.logfile=./vPblCalib.log

# Here logging can be enabled or disabled (true/false)
.loggingEnabled=true

# Here the maximum file size of the logfile can be specified
# Note: The log file size can exceed the specified limit with the size of the
log entries of the last Setup/(Reverse-)Transformer run
.maxFilesize=1000

# Here the global log level is defined (INFO/SEVERE)
.level=INFO
```

Figure 3-4 Properties of the logging mechanism

4 Integration

This chapter gives necessary information for the integration of the vPblCalib into the Cfg5 generation process as well as the calibration process.

4.1 Scope of Delivery

The delivery of the vPblCalib contains the files described in the following sections:

4.1.1 Static Files

File Name	Description
CommonAsr_vPblCalib.dll	This is the DLL that realizes the A2L transformer interface specified in ASAM MCD-2 MC version 1.7.0 [1]. It is used to incorporate vPblCalib with the calibration tool.
CommonAsr_vPblCalib.jar	This java application contains the main logic of vPblCalib and is responsible for the interaction with DaVinci Configurator 5 in background as well as generating all necessary files for calibration.

Table 4-1 Static files

4.1.2 Dynamic Files

The dynamic files are generated by the vPblCalib.

File Name	Description
vPblCalib.a2l	This is the A2L file which describes all the parameters that are available for calibration via the calibration tool. It is created during generation of the DaVinci Configurator 5 project. The file is only an A2L-file fragment and needs to be merged into the calibrators main A2L-file.
vPblCalib.cdfx	This is file contains a parameter data set of all parameters available for calibration. It is created every time the Cfg5 project is generated or the conversion to the compressed data representation is triggered by the calibration tool. It is stored internally by the vPblCalib and is used in the decompression step.

Table 4-2 Generated files

4.1.3 Template Files

File Name	Description
CommonAsr_vPblCalib.config	This configuration file needs to be placed next to CommonAsr_vPblCalib.dll. This file contains the path to the SIP. The user needs to adapt this file when using vPblCalib in Calibration phase. The SIP path must be free of spaces.
CommonAsr_vPblCalib_ExecutePB.bat	This is the batch script executed by vPblCalib to trigger the Post-Build Loadable tool chain. It contains an example configuration with Vector PB tool chain. The user needs to adapt this template file to the needs of the concrete Cfg5 project and if necessary, to the user defined PB Loadable tool chain.

Table 4-3 Template files

4.2 Integrate vPblCalib into a Cfg5 Project (Setup Phase)

To integrate vPblCalib into a Cfg5 project:

- Make sure the following pre-conditions are fulfilled:
  - > Individual Post-Build Loadable support is enabled for the MICROSAR modules which shall provide the capability for calibration
  - > The ECUC files are located in *<ProjectPath>\Config\ECUC*.ecuc.arxml*

    where *<ProjectPath>* is the path of the Cfg5 project
- Add the Automation Interface script task *vPblCalib_Setup* as additional External Generation Step in the project of Cfg5
  - > Project → Project Settings → Code Generation → External Generation Steps → Add Script Execution Step → Select the script task *vPblCalib_Setup* and press Finish

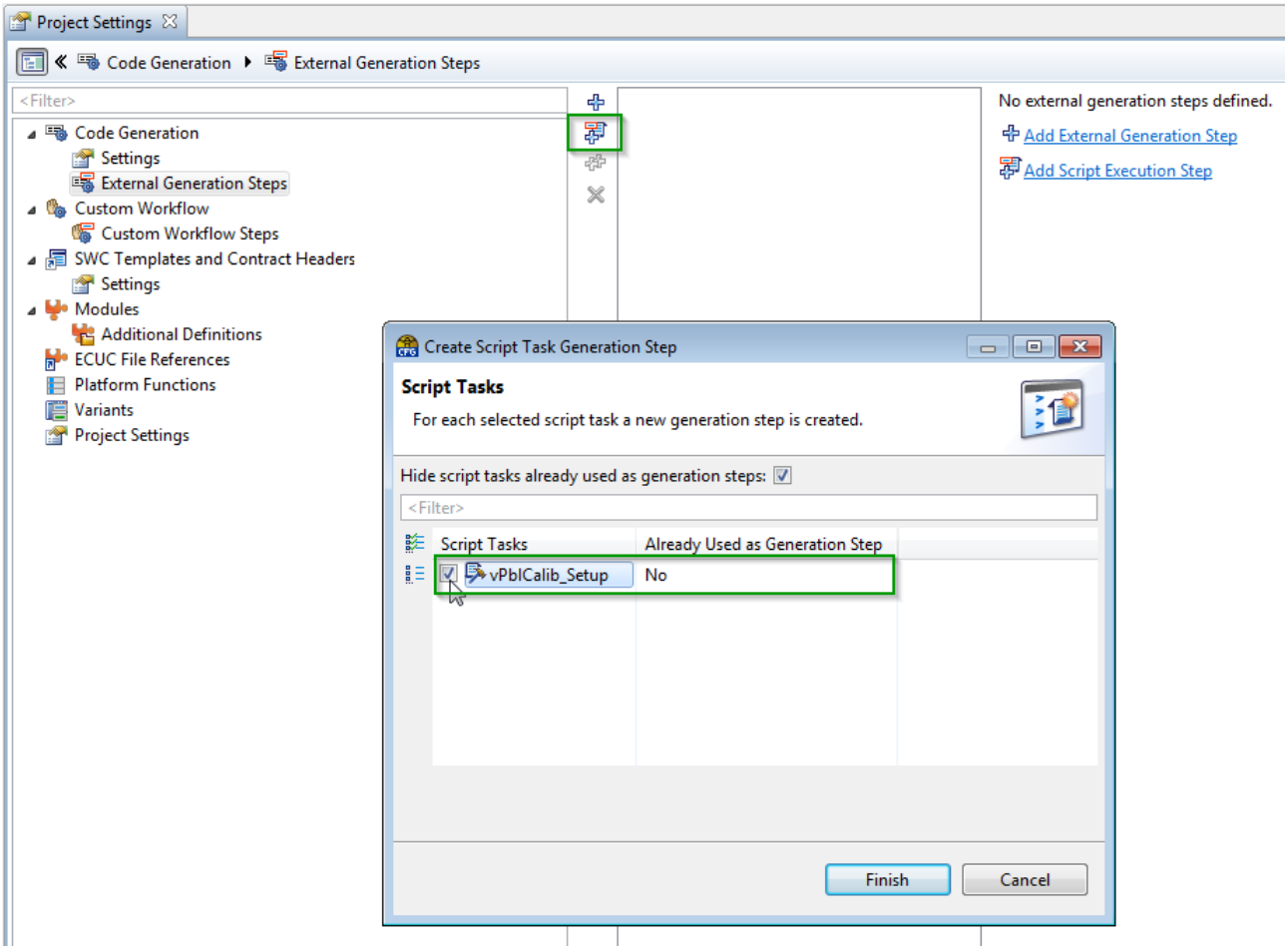


Figure 4-1 Add vPblCalib_Setup as additional External Generation Step

- Now, *vPblCalib_Setup* is available as an additional generation step in the generation view of Cfg5. Before executing the generation, make sure that the generation step *vPblCalib_Setup* is executed after the code generation of the MICROSAR modules that are to be calibrated. If necessary move the *vPblCalib_Setup* step below the generation steps of the respective modules as shown in Figure 4-2. For the later

calibration, all generators must run without reported validation messages with severity error.

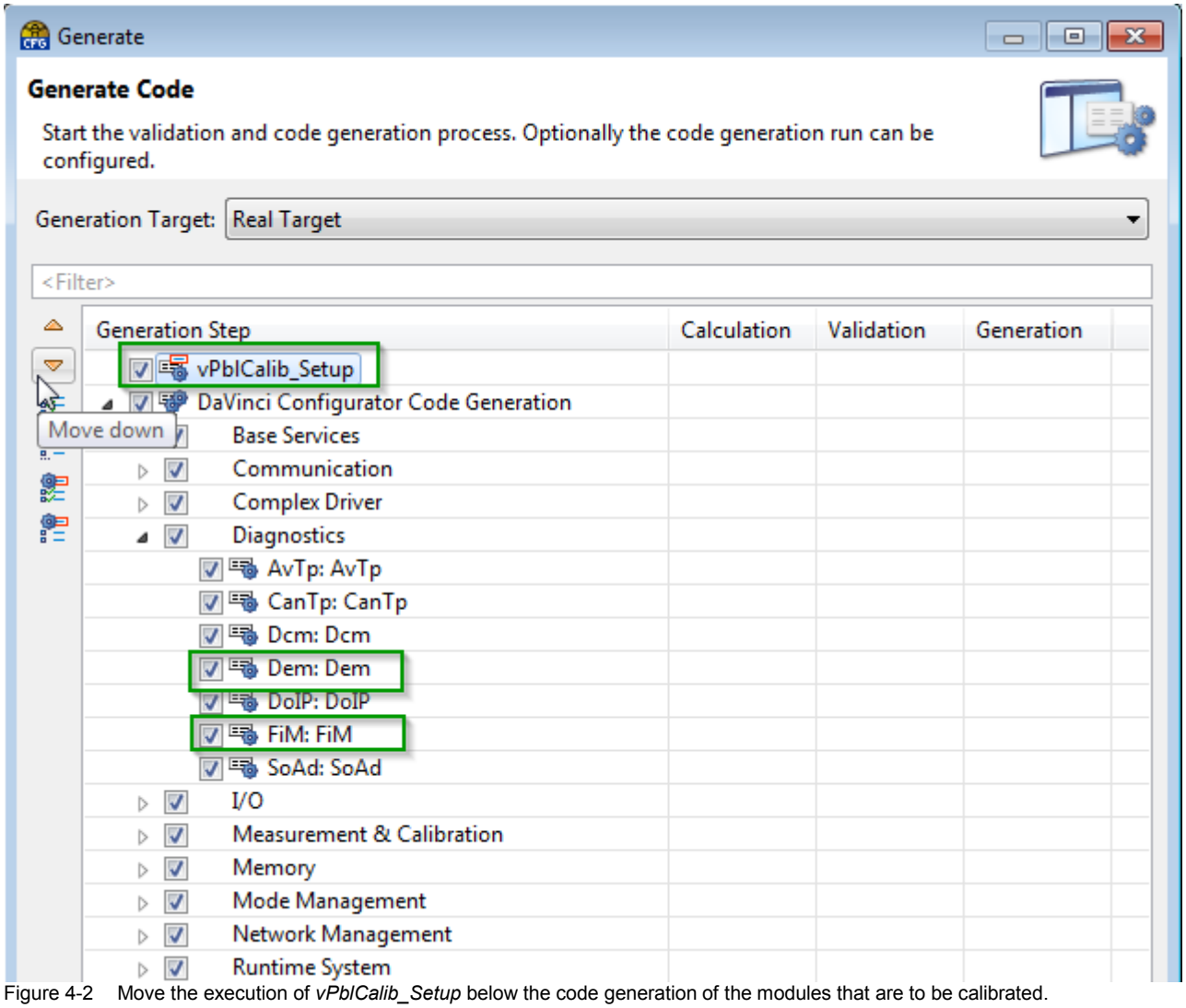


Figure 4-2 Move the execution of *vPblCalib_Setup* below the code generation of the modules that are to be calibrated.

Caution
After any change was made in the configuration of a module that is to be calibrated, *vPblCalib_Setup* needs to be run in order to track these changes for later calibration.

- Adapt the CommonAsr_vPblCalib_ExecutePB.bat file according to the needed Post-Build Loadable tool chain. It is possible to adapt this script to work with a user-defined Post-Build Loadable tool chain which provides the binary file at the specified location. For the PBL tool chain of Vector the adaption of the input parameters for PBXmlGenerator and HexView according to the concrete project needs is required. Detailed information is provided in the Technical Reference for PBXmlGenerator [5] and HexView [6]. The batch file must return the PB section of the module (defined by input parameter) as padded binary file at the specified target location (defined by input parameter).

4.3 Integrate vPblCalib into the calibration tool environment (Calibration Phase)

To integrate vPblCalib into the calibration tool environment:

- Make sure the following pre-conditions are fulfilled:
  - > Generation of *vPblCalib_Setup* succeeded in Setup Phase.
  - > The created vPblCalib.a2l file of the final generation run is merged into the main A2L file used in the calibration tool.
  - > The SIP and Cfg5 project after the final generation run is available.
  - > The execution of CommonAsr_vPblCalib_ExecutePB.bat with valid parameters succeeds.
    - > When used without modifications, this requires the tools PBXmlGenerator and HexView to be installed.
- Copy the CommonAsr_vPblCalib.dll and the CommonAsr_vPblCalib.config file next to the used main A2L file.
- Adapt the specified SIP path in CommonAsr_vPblCalib.config to match the one at the local machine.

5 API Description

This chapter contains a description of provided and required interfaces of vPblCalib. Please see Figure 2-1 for an overview of the interfaces.

5.1 Services provided by vPblCalib

5.1.1 vPblCalib_Setup

vPblCalib provides this service as Cfg5 Automation Interface script.

Script name	
`vPblCalib_Setup`	
Parameter	
N/A	N/A
Return code	
N/A	N/A
Functional Description	
vPblCalib_Setup generates the vPblCalib.a2l and the vPblCalib.cdfx file depending on the configuration of all calibratable modules. Additionally, the vPblCalib.cdfx file is stored in the internal parameter data set collection.	
Particularities and Limitations	
> Pre-condition: The configuration of all calibratable modules must be free of validation messages with severity Error.	
Expected Caller Context	
> Cfg5 triggers this AI script as separate generation step.	

Table 5-1 vPblCalib_Setup

5.1.2 A2L Transformer Interface

vPblCalib provides the following A2L Transformer DLL functions as specified in [1]:

> IsTransformationPossible

> GetTransformerDescription

> GetDllDescription

> ExecuteTransformation

These functions are called by standard calibration tools. Pre-condition for the ExecuteTransformation function is the successful execution of the *vPblCalib_Setup* generation step. As well as the correct configuration of the CommonAsr_vPblCalib.config file.

5.2 Services used by vPblCalib

5.2.1 CommonAsr_vPblCalib_ExecutePB.bat

Batch file name	
CommonAsr_vPblCalib_ExecutePB.bat	
Parameter	
SIPPath	Path to the SIP/external directory.
Cfg5ProjectPath	Path to the Cfg5 project directory.
Cfg5DpaFilePath	Path to the Cfg5 Dpa file directory.
ModuleShortName	Short name of the module.
ModulePBROMStartAddress	The PB ROM start address of the module defined by /MICROSAR/EcuC/EcucGeneral/PostbuildLoadable/IndividualPostBuildLoadableModule/ModulePBROMStartAddress.
ModulePBROMEndAddress	The PB ROM end address of the module defined by /MICROSAR/EcuC/EcucGeneral/PostbuildLoadable/IndividualPostBuildLoadableModule/ModulePBROMEndAddress.
FillByte	Specified fill byte.
OutputFile	The path where the resulting binary file is expected.
Return code	
N/A	N/A
Functional Description	
CommonAsr_vPblCalib_ExecutePB.bat triggers the PB-Loadable tool chain and returns the PB section as padded binary.	
Particularities and Limitations	
> Post-condition: The resulting PB section is padded with the fill byte and has size=ModulePBROMEndAddress – ModulePBROMStartAddress.	
Expected Caller Context	
> vPblCalib triggers this batch script during calibration phase for getting the compressed data representation (PB section).	

Table 5-2 CommonAsr_vPblCalib_ExecutePB.bat

5.2.2 Automation Interface

vPblCalib internally uses the Cfg5 Automation Interface to update the Cfg5 project and to re-generate the MICROSAR BSW modules. More information about the Automation Interface is provided in [4].

6 Configuration

6.1 Configuration Variants

vPblCalib supports the configuration variants

- VARIANT-POST-BUILD-LOADABLE

6.2 Configuration Attributes

The description of each configurable attribute is available as online help in the DaVinci Configurator Pro 5 tool.

The parameters in Table 6-1 can be configured via the tool.

Configuration Parameter	Comment
/MICROSAR/vSet/vSetCalibration/ vSetPBCalibrationVirtualStartAddress	Start address of the virtual memory segment. The address should not lie in the address range of the ECU. This parameter is mandatory for the usage of vPblCalib. **Recommendation**: Choose an address beyond the ECU's address range. The virtual-memory segment size depends on the concrete configuration and is calculated by vPblCalib. **Note**: vPblCalib does not support the VTT capability.

Table 6-1 Configurable Parameters



Note
Due to configuration parameter “vSetPBCalibrationVirtualStartAddress”. The MICROSAR vSet module is mandatory for the usage of vPblCalib.

7 Abbreviations

Abbreviation	Description
AI	Automation Interface
API	Application Programming Interface
ASAM	Association for Standardization of Automation and Measuring Systems
BSW	Basis Software
CDF	Calibration Data Format
Cfg5	DaVinci Configurator 5
DEM	Diagnostic Event Manager
DLL	Dynamic Link Library
ECU	Electronic Control Unit
FiM	Function Inhibition Manager
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
PB	Post-Build
PBL	Post-Build Loadable
SIP	Software Integration Package
VTT	vVIRTUALtarget

Table 7-1 Abbreviations

8 Contact

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